

ARIMA Time Series Forecasting for Research

A 60-minute workshop for CCIS faculty

Oct 2, 2026 | 3 PM to 4 PM

Slides, notebooks, and web explorer: github.com/ahleksu/arima-workshop-ccis

Today in 60 minutes

Time	Segment	You will leave with
0 to 5	Why ARIMA in research	A rule for when it fits
5 to 15	Stationarity and differencing	A test you can defend
15 to 25	AR, MA, ACF, PACF	How to read the plots
25 to 42	Fit ARIMA and SARIMAX	A model you can diagnose
42 to 53	Forecast and validate	An honest evaluation
53 to 60	Take-home	A plan for your own data

Where ARIMA shows up in CCIS work

- **Systems:** server load, API calls, network traffic, storage growth
- **Software engineering:** issue arrival, build failures, commit activity
- **Education:** enrollment, course demand, library use, exam attempts
- **IoT and energy:** sensor streams, power use, temperature
- **Information systems:** sales, help-desk tickets, app usage
- **Dissertations:** the **baseline** you compare an LSTM or Prophet model against

When ARIMA fits, and when it does not

Good fit

- One series, regular spacing
- 50 to a few thousand points
- Patterns you can describe: trend, season, memory
- You need intervals and a model you can explain

Poor fit

- Many related series, or huge feature sets
- Strong non-linear or regime-switching behavior
- Very irregular or event-driven data
- Very long seasons (365 with daily data)

The workflow in one slide

1. **Plot** the data and read it
2. **Stabilize**: transform and difference until stationary
3. **Identify**: ACF and PACF suggest orders
4. **Estimate**: maximum likelihood
5. **Diagnose**: residuals must look like white noise
6. **Forecast** with intervals
7. **Validate**: backtest against a baseline

Box and Jenkins, 1970. Steps 3 to 5 repeat until the model is adequate.

Part 1

Time series fundamentals

Notebook `01_fundamentals_stationarity`

Four parts of a series

- **Trend:** slow movement up or down
- **Seasonality:** a fixed repeating pattern (week, year)
- **Cycle:** irregular long swings
- **Noise:** what is left

Our data: daily electricity demand, 2015 to 2024. It is **synthetic**, so we know the truth.

Weak stationarity

A series y_t is weakly stationary when

1. $E[y_t] = \mu$ (constant mean)
2. $\text{Var}(y_t) = \sigma^2 < \infty$ (constant variance)
3. $\text{Cov}(y_t, y_{t+h}) = \gamma(h)$ (depends only on the lag)

Why we care: ARIMA estimates **one** set of coefficients. That only makes sense if the process does not change over time.

Test it: Augmented Dickey-Fuller

- Null hypothesis: **unit root** (non-stationary)
- Small p-value (below 0.05): reject, evidence of stationarity
- Pair it with **KPSS**, which has the opposite null

Live result on our data

- Level, constant only: $p = 0.63$. Cannot reject.
- Level, constant and trend: $p < 0.001$. Rejects. **Trend-stationary.**
- After one difference: $p < 0.001$.

Fix it: difference and transform

- **First difference** $y_t - y_{t-1}$ removes a stochastic trend
- **Seasonal difference** $y_t - y_{t-7}$ removes a weekly pattern
- **Box-Cox** stabilizes variance when swings grow with the level
- Difference as **little** as possible. Over-differencing adds noise.

Invert the transform after forecasting, and report errors on the original scale.

Part 2

AR and MA processes

Notebook `02_ar_ma_acf_pacf` and the **web explorer**

Autoregression and moving average

AR(p): today depends on its own past

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

MA(q): today depends on recent shocks

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

AR needs $|\phi| < 1$ (stationary). MA needs $|\theta| < 1$ (invertible).

Reading the ACF and PACF

Process	ACF	PACF
AR(p)	Decays	Cuts off after lag p
MA(q)	Cuts off after lag q	Decays
ARMA(p, q)	Decays	Decays

The blue band is about $\pm 1.96 / \sqrt{n}$. One in twenty spikes crosses it by chance.

Try it yourself: three mystery series

Notebook 02, section 6.

- Look at the ACF and PACF of series A, B, and C
- Write down your guess
- Run the reveal cell

Lesson from the live run: the criteria do not always find the truth. For A, both picked ARMA(2,2) and the truth is AR(2). For B, AIC picked ARMA(2,2) and BIC picked the true MA(1). For C, both were right. Report both criteria, check whether the extra terms are significant, and prefer the simpler model when scores are close.

Part 3

ARIMA and SARIMAX on energy demand

Notebooks 03 and 04

ARIMA(p, d, q)

$$\phi(B) (1 - B)^d y_t = c + \theta(B) \varepsilon_t$$

- **p** autoregressive terms
- **d** number of differences
- **q** moving average terms
- Estimated by **maximum likelihood**: the coefficients that make the data most probable

Select with **AIC** and **BIC**. Lower is better. Within 2 points, take the simpler model.

Live: fit and diagnose (notebook 03)

1. Hold out the last 26 weeks
2. ADF says $d = 1$
3. Grid search $p, q \in \{0..3\}$
4. Parsimony rule picks **ARIMA(0,1,1)**
5. **Ljung-Box** on residuals: passes at lags 10 and 26, borderline at lag 52
6. Forecast is **flat**, MAE about 18.5 GWh (1.1 percent)

A low AIC is not proof that the model is adequate.

Seasonality and outside drivers

$\text{ARIMA}(p, d, q)(P, D, Q)_s$ $s = 7$ for daily data with a weekly pattern

SARIMAX adds regressors: temperature (heating and cooling degrees) and holidays.

Model	MAE, 6-month test
SARIMA, no drivers	23.0 GWh
SARIMAX, with drivers	15.2 GWh

Estimated effects: 6.5 GWh per heating degree, 5.9 per cooling degree, and -146 GWh on a holiday.
The generator used 6.5, 6.5, and -140.

Three traps

1. **Multicollinearity:** temperature with heating and cooling degrees is perfectly collinear. Check VIF.
2. **Future regressors:** you need their future values to forecast.
3. **Over-differencing:** the seasonal MA coefficient near -1 is a warning. For fixed calendar patterns, try weekday dummies instead.

`pmdarima.auto_arima` is a good first pass. Still diagnose.

Part 4

Forecast and validate

Notebook `05_forecasting_validation`

Point forecast and interval

- A point forecast hides the uncertainty
- A 95 percent interval should cover about 95 percent of actual values
- In our test, coverage was **98.9 percent**. The intervals were too wide because six outliers inflated the error variance.

Always check coverage. Report it.

Metrics and a baseline

Metric	Use
MAE	Same unit as the data
RMSE	Punishes large errors
MAPE	Percent. Breaks near zero
MASE	Below 1 beats the seasonal naive baseline

Rolling-origin backtest (26 origins, 7-day horizon):

Method	MAE	MASE
SARIMAX, expanding window	12.9	0.47
SARIMAX, rolling 730 days	13.2	0.48
Seasonal naive	26.4	0.95

Breaks and outliers

- A demand shock starts on 2020-03-15. A model fitted before it cannot see it: **MAE 81 GWh** during the shock.
- Handle a break: **detect and report, intervention variable, shorter window, wider intervals.**
- Outliers: flag with standardized residuals. Each spike leaves a smaller **echo** the next day.
- Find out **why** before you act. Never delete points silently.

From notebook to pipeline

1. **Validate** the input. Fail loudly.
2. **Fit** and **forecast**
3. **Record** data range, orders, versions
4. **Monitor** recent error. Alert past a threshold
5. **Fallback** to the seasonal naive forecast

A forecast that runs once is a notebook. One that runs every week is a pipeline.

Take-home

Make it yours

Reviewing a forecasting chapter: 8 questions

1. Is the data source, frequency, and period stated?
2. Were breaks and outliers found and explained?
3. How was stationarity decided, and which tests?
4. How were orders chosen? Is the grid stated?
5. Do the residual checks pass?
6. Is there a baseline, and does the model beat it?
7. Is the evaluation out of sample, from many origins?
8. Are intervals reported, with their coverage?

Your next hour

1. Open **notebook 06**. Run it once on the sample.
2. Change `DATA_PATH` and four settings. Run it on your series.
3. Read the data checklist output before you trust anything.
4. Copy the methods paragraph and edit it.

Privacy: keep sensitive data off Colab. Use the local setup script.

Where to go next

- *Forecasting: Principles and Practice* (3rd edition), free online: otexts.com/fpp3
- statsmodels time series docs: statsmodels.org/stable/tsa.html
- `pmdarima` docs: alkaline-ml.com/pmdarima
- Extensions: ARIMAX with lagged drivers, state-space models, ETS, hierarchical forecasting, probabilistic metrics

Everything from today: github.com/ahleksu/arima-workshop-ccis

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Questions

Come back to the series you wrote on the board at the start.